

Functions & Graphs

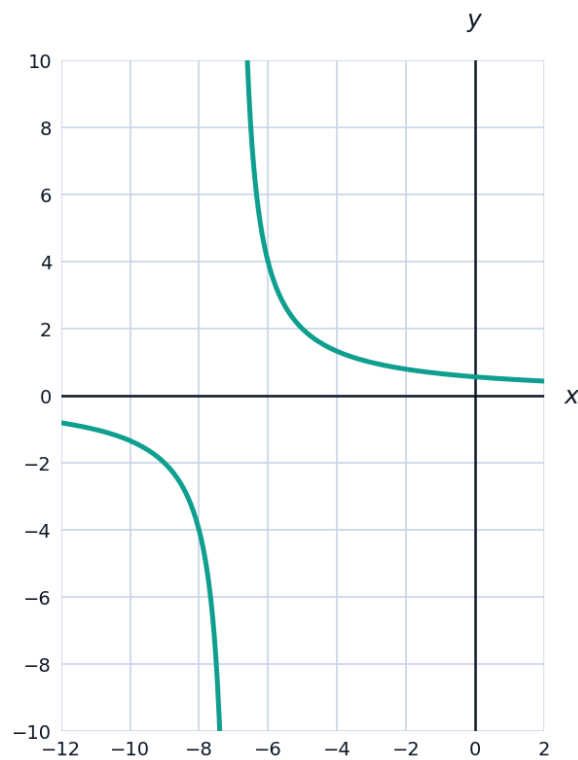


Evaluating functions

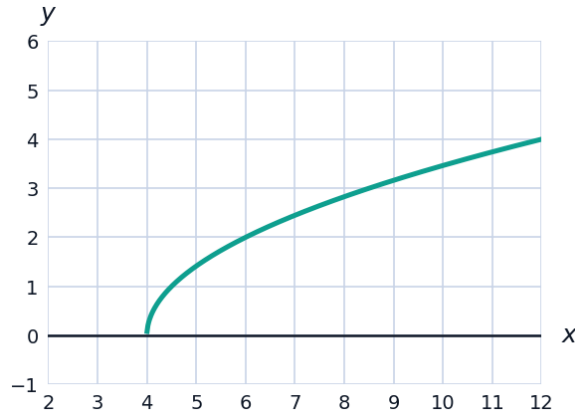
- 1 If $f(x) = 3x^2 - 2x + 5$, find $f(-2)$.
- 2 If $g(x) = \frac{2x+1}{x-3}$, find $g(5)$.

Domain and range

- 3 Find the domain of $f(x) = \frac{4}{x+7}$.



- 4 Find the domain of $g(x) = \sqrt{2x - 8}$.

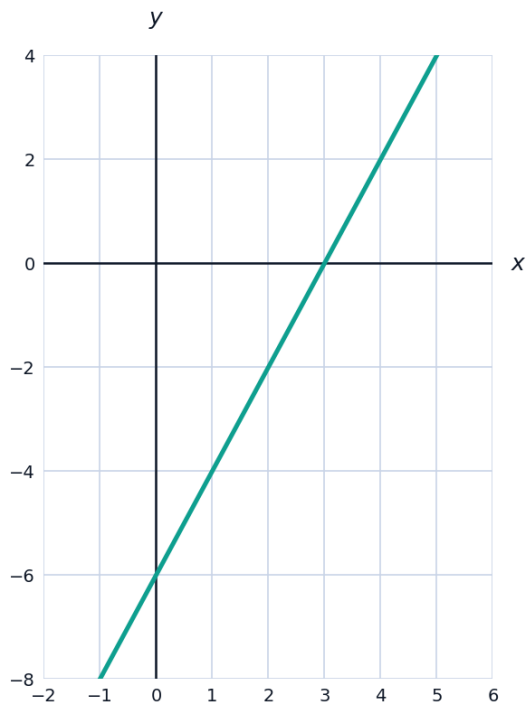


Composite functions

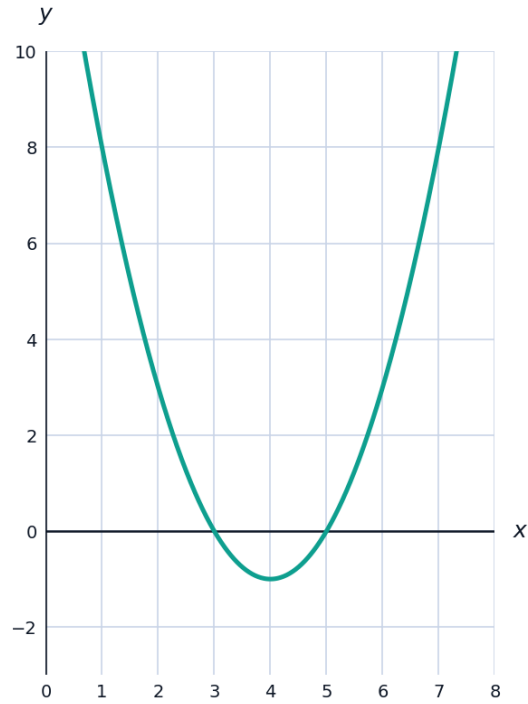
- 5 If $f(x) = 2x + 1$ and $g(x) = x^2 - 3$, find $f(g(2))$.
- 6 If $f(x) = x + 4$ and $g(x) = 3x$, find $(f \circ g)(x)$ and $(g \circ f)(x)$, and explain why they differ.

Graphing linear and quadratic functions

- 7 Find the x- and y-intercepts of $y = 2x - 6$.

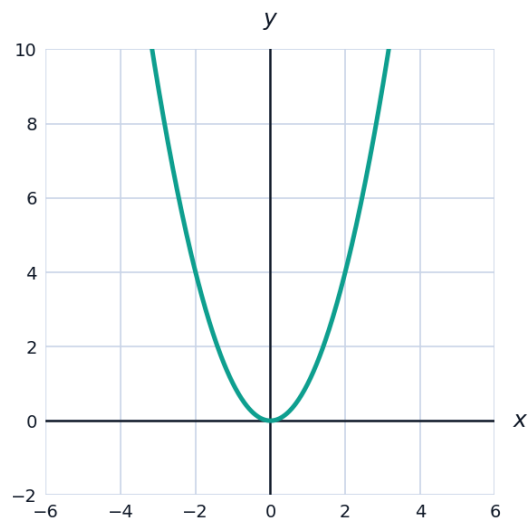


- 8 Find the vertex of $y = x^2 - 8x + 15$.

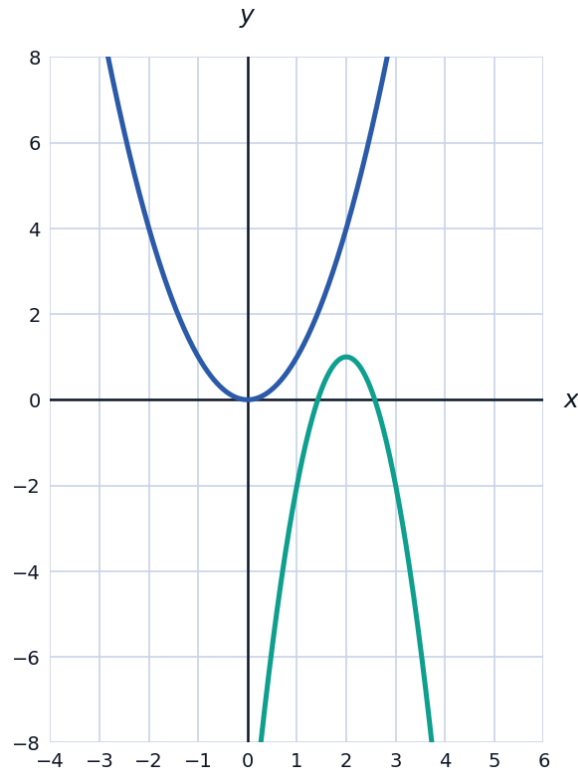


Transformations of graphs

- 9 The graph of $y = x^2$ is shifted 4 units left and 3 units down. Write the equation of the resulting graph.

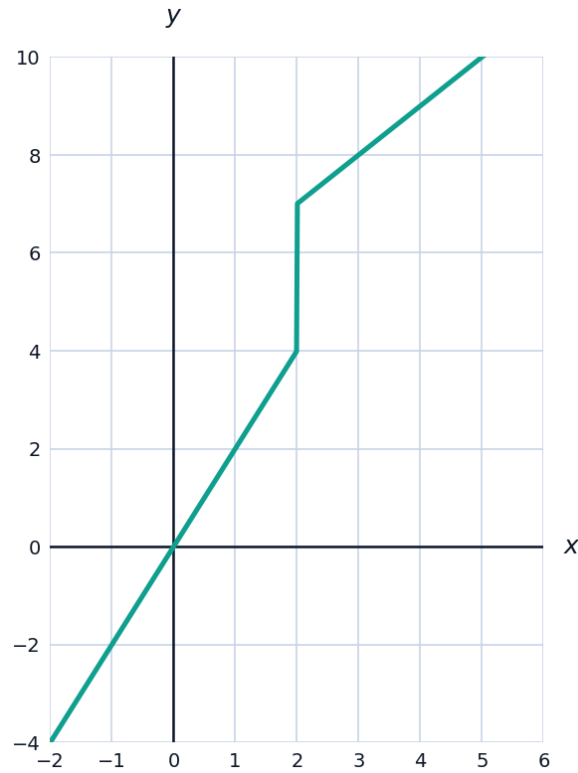


- 10 Describe how the graph of $y = -3(x - 2)^2 + 1$ differs from the graph of $y = x^2$.



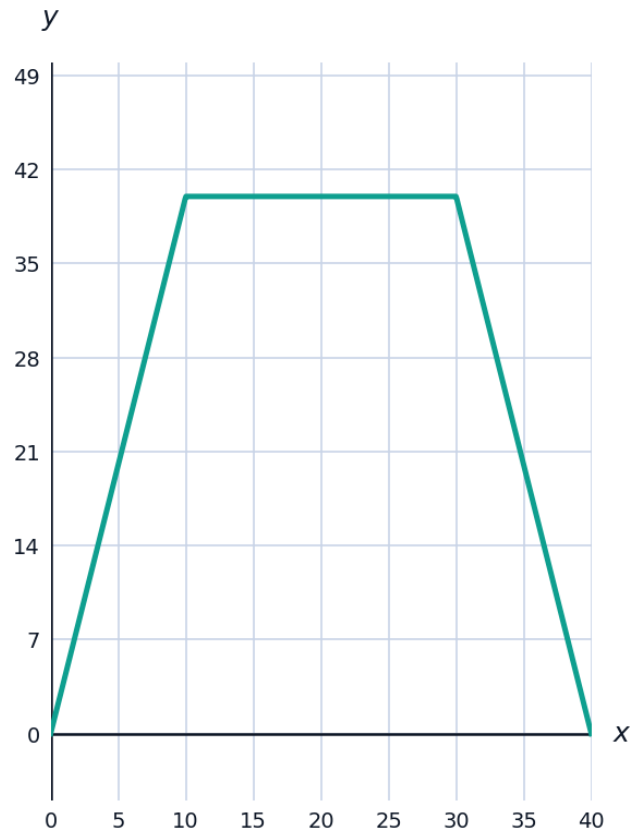
Piecewise and step functions

- 11 A function is defined as $f(x) = 2x$ for $x \leq 2$ and $f(x) = x + 5$ for $x > 2$. Find $f(0)$ and $f(4)$.



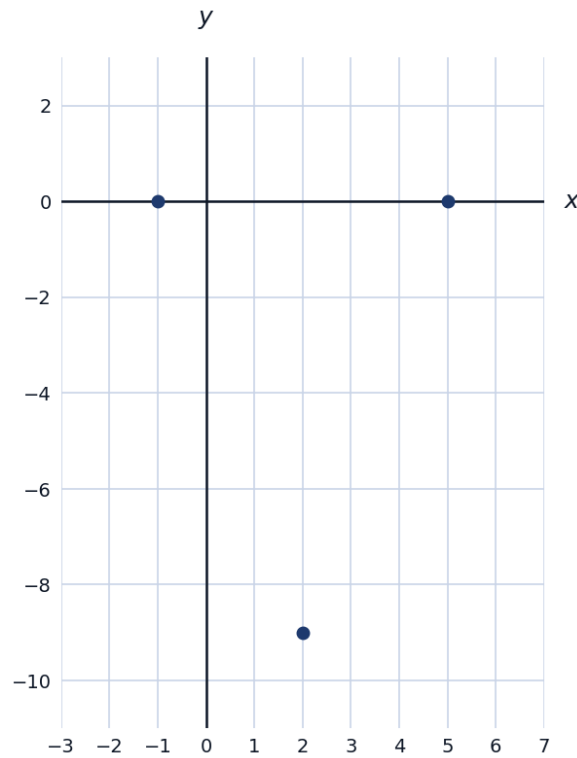
Interpreting graphs in context

- 12** A graph shows a car's speed over time: the speed increases from 0 to 40 mph over the first 10 seconds, stays constant at 40 mph for the next 20 seconds, then decreases to 0 over the final 10 seconds. Describe what is happening to the car during each phase.



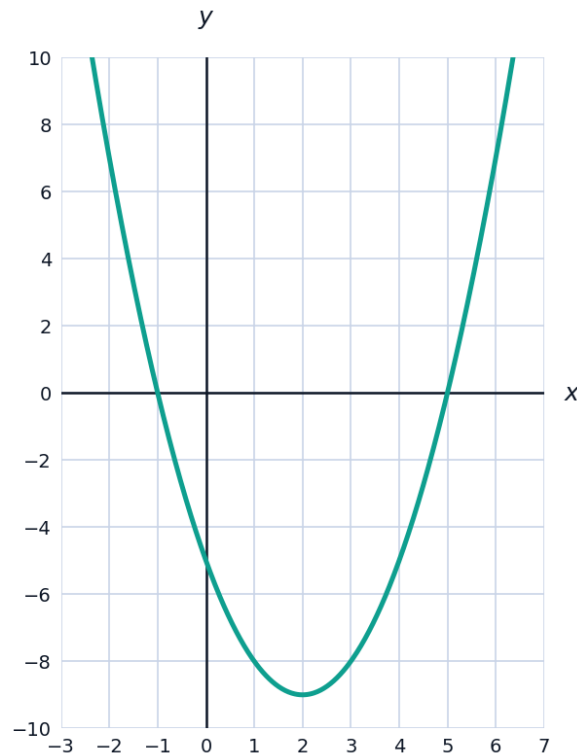
Finding the equation of a function from a graph description

- 13 A parabola has x -intercepts at $x = -1$ and $x = 5$, and passes through $(2, -9)$. Find its equation in factored form.



Multi-part function analysis

- 14** This question has two parts. A function is given by $f(x) = x^2 - 4x - 5$. (a) Find the zeros of the function by factoring. (b) Find the vertex, and state whether it is a minimum or maximum.

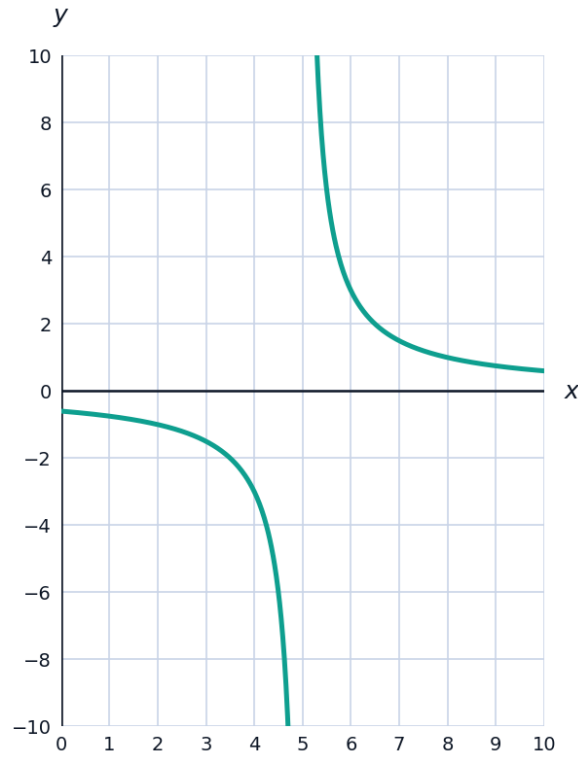


Inverse functions

- 15** Find the inverse of $f(x) = 4x + 8$.
- 16** If $f(3) = 10$, find $f^{-1}(10)$, using the definition of an inverse function.

Rational and radical functions

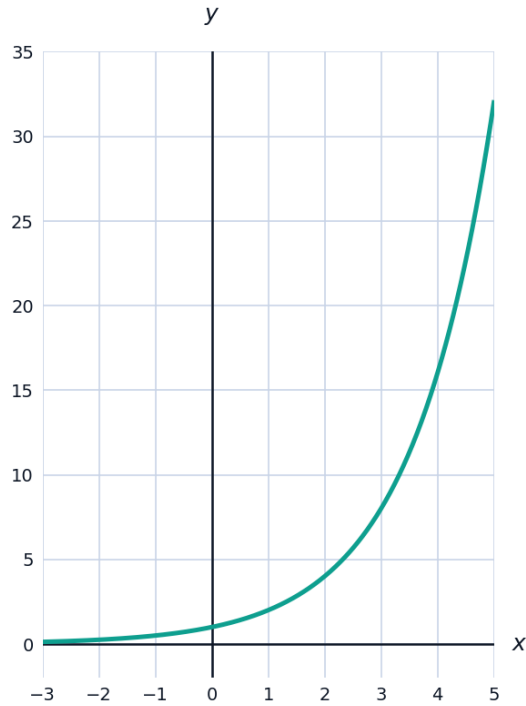
- 17 Find the vertical asymptote of $f(x) = \frac{3}{x-5}$.



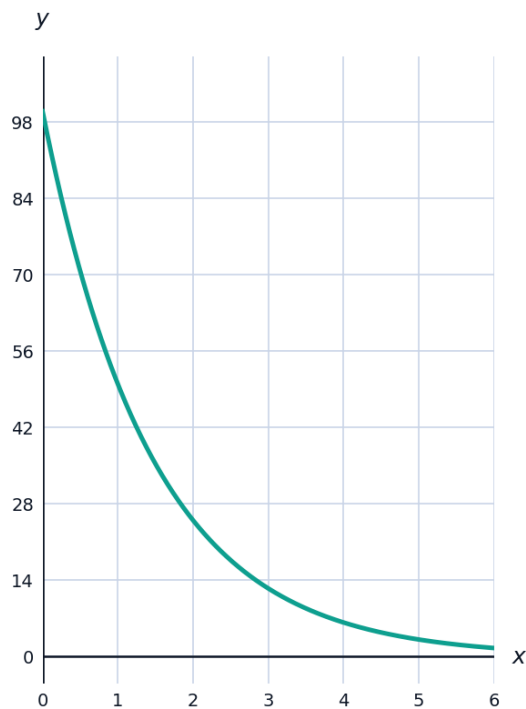
- 18 Find $g(9)$ for $g(x) = \sqrt{x} + 4$.

Exponential functions

- 19 If $f(x) = 2^x$, find $f(5)$ and $f(-2)$.



- 20 A function is given by $g(x) = 100(0.5)^x$. Find $g(3)$ and interpret its meaning if this models a decaying quantity over x time periods.



Multi-part analysis of a real-world function model

- 21 This question has two parts. A ball is thrown, and its height is modeled by $h(t) = -16t^2 + 48t + 4$, where t is time in seconds. (a) Find the height of the ball at $t = 0$ and interpret this value. (b) Find the time at which the ball reaches its maximum height, and find that maximum height.

